TACT: Trust-Anchored Confidence Tempering for Self-Consistency Voting in Large Language Models

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Abstract—Confidence-weighted self-consistency improves on majority voting when a frozen model's self-reported confidence is calibrated in direction. Every published scheme is monotone increasing in confidence, so an anti-correlated channel poisons the vote, and binary calibration gates survive inversion only by discarding discriminative signal. tact derives the vote exponent from the measured, signed, within-item discrimination of the channel: a pooled van Elteren Somers' D with an item-clustered standard error, positive-part James–Stein shrinkage, and a Bayes-discriminant link, which at the default base rate $\bar{p} = \frac{1}{2}$ collapses to $\gamma = z\sqrt{2+z^2}$ with z the probit of the shrunk pooled AUC. Inside the shrinkage dead zone the vote is bit-identical to plain self-consistency. A label-free variant estimates the sign from agreement pseudo-labels under an attenuation identity that guarantees sign consistency while the plurality-error rate stays below one half; past that boundary it mis-signs, which the paper measures and reports. On a synthetic-oracle harness it recovers anti-correlated channels that pin every published protocol to the majority floor (1.000 vs. 0.762) and cracks the heterogeneity floor with zero paired losses (0.923 vs. 0.785). Against a dev-picked signed grid, which the sweep shows is a far stronger baseline than the published protocols, the advantage narrows to distortion, echo, and label-free operation. Two real-trace campaigns then bound the setting: the channel is null on saturated benchmarks ($z = -1.24$) and positive on competition mathematics ($\widehat{D} = +0.250$, $z = +2.54$), yet the stratum any such method can act on measures 2.5–7.5% of items across five substrates in two domains, so abstention is the correct default and the dead zone implements it.

Index Terms—large language models, self-consistency, confidence calibration, label-free estimation, rank statistics

I. Introduction

Self-consistency (sc) [1] improves the reasoning accuracy of a frozen large language model (LLM) by sampling K chain-of-thought traces and returning the plurality answer. Each trace can also report a confidence score, verbalized [2], [3], derived from token log-probabilities, or elicited as $P(\text{True})$ [4], so weighting votes by confidence is a natural refinement. Confidence-Informed Self-Consistency (CISC) [5] showed that this recovers the accuracy of plain sc at a fraction of the sampling budget, and introduced Within-Question Discrimination (WQD) to argue that discrimination, not calibration, is the property that makes a confidence signal useful for voting.

This refinement carries a structural fragility that, to the best of our knowledge, no published method addresses. Every existing weighting scheme is monotone increasing in confidence, including CISC's softmax weights, reliability-aware pseudo-counts [6], and warmup-thresholded filtering [7]. The trust decision is which magnitude of up-weighting to apply; the possibility that the channel is anti-correlated with correctness is never searched for. Yet miscalibration of direction is not exotic: reinforcement fine-tuning is known to distort verbalized confidence, distribution shift can invert a signal that was informative in-domain, and in the experiments reported here a simple anti-correlated channel ($\kappa = -0.6$; Section III) drives confidence-weighted baselines from near-perfect accuracy to far below the majority-vote floor, while the same evidence, read with the correct sign, is a perfect signal. The defensive alternative, a binary dev-set gate that disables the channel when calibration error is high, survives the inversion but discards discriminative signal wholesale: a systematically under-confident yet perfectly ranked channel fails an ECE gate for reasons irrelevant to voting utility [5], [8].

This paper frames the problem as estimating one scalar: the signed within-item discrimination of the confidence channel, and mapping that scalar, with its uncertainty, to a vote exponent. The contributions are:

C1: Signed, analytically-tempered weighting. tact votes with $w_i = \exp(\gamma\varphi_i)$, where φ_i is the standardized van der Waerden score of trace i's within-item confidence midrank and γ is derived, not grid-searched: a pooled van Elteren Somers' D (equal to $2 \cdot \text{WQD} - 1$) with an exact tie-corrected null variance and an item-clustered jackknife standard error, shrunk by positive-part James–Stein with a significance floor, then mapped through a Bayes-discriminant link. Two anchors are exact: inside the dead zone the vote is bit-identical to sc via a shared code path, and the log-value feature map reproduces CISC-power. Since φ uses only within-item ranks, the method is invariant to every strictly monotone distortion of the confidence scale, and under compression it beats the best exponent in the tested grid $\{0.25, 0.5, 1, 2, 4\}$ (1.000 vs. 0.963; that optimum sits at the largest exponent tried, so the comparison bounds the grid, not the family).

C2: Label-free estimation of the sign. The crowdsourcing lineage estimates reliability from cross-annotator covariance [9], [10]; one exchangeable channel from one model offers no such structure. tact estimates the signed discrimination from deduplication-weighted agreement pseudo-labels under a proven class-conditional-noise identity, $\mathbb{E}[\widehat{D}_g] = (1 - 2\bar{\rho})D$: while the pair-weighted plurality-error rate $\bar{\rho}$ is below $1/2$ the estimate can only under-trust, never mis-sign. A split-half inversion de-attenuates conservatively and sign-aware alarms return the method to sc at the identifiability boundary. Past $\bar{\rho} = 1/2$ it does mis-sign, which Section IX measures rather than assumes away.

C3: An impossibility result and its structured escape. With i.i.d. per-item coupling and no observable covariate, per-item label-free adaptation is closed: any monotone use of an item's own agreement statistic collapses to plurality reinforcement, the observable sign opposes the truth on 96% of the plurality-wrong items with $|D_q| > 0.3$, the ones where a flip could win, and $\{\kappa > 0, \text{minority right}\}$ and $\{\kappa < 0, \text{plurality right}\}$ induce the same observable law. Indexed instead by an observable covariate, the same estimator run per group recovers each group's signed coupling with zero paired losses to sc (Section VI).

C4: A pre-registered falsification protocol. Four falsifiers were fixed before implementation, two of them designed to kill the method: the published dev-calibrated CISC protocol, whose tuned temperature already interpolates sc$\leftrightarrow$CISC, and a dev-picked signed exponent grid, which the sweep shows is far the stronger of the two. All four survived, and the margins are reported both ways: against the signed grid the advantage concentrates in distortion, echo, and label-free operation, and tact trails it in the mid-range.

C5: A measurement of the addressable stratum. Two real-trace campaigns and a five-substrate window measurement bound what any label-free aggregation method can do. The channel is null on saturated benchmarks ($\widehat{D} = -0.219$, $z = -1.24$) and positive on competition mathematics ($+0.250$, $z = +2.54$), so the premise holds where the model is uncertain; but the stratum such a method can act on is 2.5–7.5% of items on every substrate tried, in two domains, and does not widen as items harden. On both real substrates the in-pool oracle itself cannot clear the pre-registered endpoint, which makes abstention the correct default rather than a conservative one.

II. Related Work

Confidence-weighted self-consistency. sc [1] treats sampled traces as i.i.d. votes. CISC [5] weights votes by softmax-normalized confidence with a temperature tuned on a labeled split, and its WQD metric makes the discrimination-vs-calibration point that also motivates this work; the rank-calibration line [8] reaches the same conclusion independently. Weighted variants [11] and early-stopping families [12], [13] refine the budget. Self-certainty [14] is the closest relative in spirit, being the one published selector that scores candidates by a rank-like quantity rather than by raw confidence, but it ranks across candidates with a fixed positive orientation and is not evaluated here; reliability-aware pseudo-counts [6] and warmup-thresholded filtering [7] adapt online but only re-scale positive trust. None of these searches a negative exponent: the obstruction is a sign bit in the hyperparameter grid rather than the weight family itself, as this paper's own SignGrid-dev baseline shows by opening the same c^{γ} family to negative γ and reaching the signed oracle across the negative half-axis. What no published protocol does is estimate that sign, with or without labels. The dev-calibrated variant must therefore be positioned honestly: CISC's tuned temperature is already a dev-calibrated sc$\leftrightarrow$CISC interpolation, so the novelty of tact-dev lies in the sign, the rank invariance, and the analytic (grid-free) map, not in dev calibration itself.

Reliability estimation without labels. Estimating worker reliability from agreement is classical [9], [15], [16]; spectral meta-learners [10] and recent LLM ensemble work [17], [18] exploit covariance across multiple predictors. The setting here differs: one exchangeable channel from one model, per-item vote structure, and the known failure of agreement proxies under correlated errors—met here with a quantified attenuation identity, conservative de-attenuation, and alarms in place of an unconditional claim.

Shrinkage and rank statistics. The estimator assembles classical parts: stratified rank statistics [19], the James–Stein positive-part estimator [20], effective-sample-size corrections [21], [22], and normal-scores discriminant analysis. The claim is the assembly and its anchors, not the parts.

Honest sibling result. A preceding system in the same line of work (RLEV-VoI, redundancy-discounted voting with value-of-information stopping) was evaluated under the same falsification discipline and failed it, dominated everywhere by a simple deduplication baseline, and is reported as a negative result. Its post-mortem isolated the confidence dilemma studied here.

III. Problem Setup

A. Notation

Items $q = 1, \ldots, Q$; item q has m_q sampled traces. Trace (q, i) yields an answer $a_{q,i}$ in a discrete set and a confidence $c_{q,i} \in (0, 1)$; correctness is $y_{q,i} = \mathbf{1}[a_{q,i} = a^*_q]$, unobserved at test time. Plain sc returns $\arg\max_A n_q(A)$ where $n_q(A)$ counts votes for answer A. CISC-power weights votes by $c^{\gamma}_{q,i}$ with a fixed $\gamma > 0$.

B. The confidence dilemma

The synthetic oracle draws, per item, traces from a cluster mixture with a latent correct answer and generates confidence as

$$c_{q,i} = \mathrm{clip}\big(\tfrac{1}{2} + \kappa\,(y_{q,i} - \tfrac{1}{2}) + \varepsilon_{q,i},\, 0.01,\, 0.99\big), \qquad (1)$$

with noise $\varepsilon \sim \mathcal{N}(0, 0.1^2)$ and coupling $\kappa \in [-0.6, 0.6]$. Fig. 1 maps the baseline landscape before the proposed method existed: unconditional weighting (CISC, $\gamma = 1$) collapses on $\kappa < 0$; an ECE gate never opens off the well-calibrated diagonal; a sign-corrected AUC gate over dev labels nearly saturates the homogeneous sweep. This pre-measurement fixes where a new method can legitimately claim wins—monotone distortion of the confidence scale, covariate heterogeneity, small dev sets, and label-free operation—and the evaluation holds itself to exactly those cells.

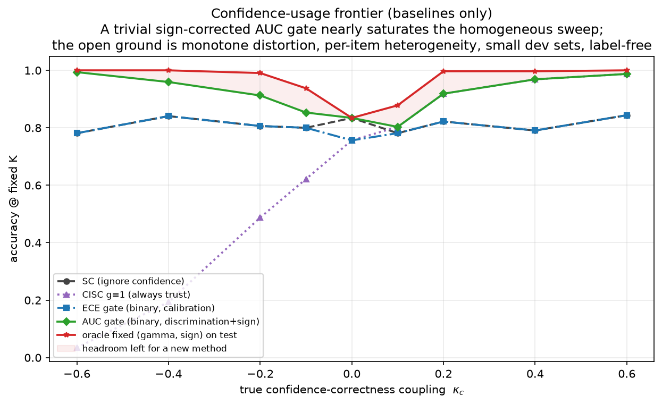

Fig. 1. The pre-measured problem statement: accuracy of baseline confidence policies at fixed K=15 as the true coupling κ varies. A trivial sign-corrected AUC gate (green) nearly saturates the homogeneous sweep; the headroom for any new method (shaded) concentrates in the mid-range and, off this plot, in distortion, heterogeneity, and label-free cells.

IV. TACT

A. Vote family

Within item q, let $R_{q,i}$ be the midrank of $c_{q,i}$ (ties averaged) and

$$\varphi_{q,i} = \frac{v_{q,i} - \bar{v}_q}{\sigma_q}, \qquad v_{q,i} = \Phi^{-1}\Big(\frac{R_{q,i}}{m_q + 1}\Big), \tag{2}$$

where σ_q is the realized standard deviation of v within the item (the no-tie value is 0.62 at m=4 but 0.95 at m=40; a closed form would silently rescale γ across budgets), and $\varphi \equiv 0$ if $\sigma_q \le 10^{-8}$ (all-tied confidences vote as plain sc). The vote is

$$\hat{a}_q = \arg\max_A \sum_{i:\, a_{q,i}=A} \exp\big(\gamma\, \varphi_{q,i}\big), \tag{3}$$

and when $\gamma = 0$ the implementation calls the sc routine itself, making the zero-trust anchor bitwise exact rather than equal in distribution. Because (2) depends on c only through within-item ranks, every strictly monotone distortion of the confidence scale leaves (3) unchanged.

B. Reliability statistic

For item q with n_q^1 positive and n_q^0 negative labels (dev: y; label-free: the pseudo-label of Section V), the Mann–Whitney statistic on midranks gives

$$D_q = 2\,\mathrm{AUC}_q - 1, \qquad \mathrm{AUC}_q = \frac{U_q}{n_q^1 n_q^0}, \tag{4}$$

which equals $2 \cdot \mathrm{WQD}_q - 1$ in CISC's notation. Pooling uses van Elteren pair-count weights $N_q = n_q^1 n_q^0$ [19]:

$$\widehat{D} = \frac{\sum_q N_q D_q}{\sum_q N_q}. \tag{5}$$

Under the within-item exchangeability null, U_q has the exact tie-corrected variance $n_q^1 n_q^0 (m_q{+}1)/12 \cdot [1 - \sum_t (t^3 - t)/(m_q^3 - m_q)]$, yielding a null standard error SE_0; between-item heterogeneity is captured by the closed-form delete-one-item jackknife SE_J. The conservative choice is

$$\mathrm{SE} = \max\big(\mathrm{SE}_0,\ \mathrm{SE}_J,\ \tfrac{1}{2\sqrt{N}}\big), \qquad r = \widehat{D}/\mathrm{SE}. \tag{6}$$

Because D is a pairwise functional, $\mathbb{E}[\widehat{D}]$ does not depend on m_q: an exponent estimated at m=40 transfers to deployment at m=8.

C. Tempering map

Shrinkage. Positive-part James–Stein with a significance floor ν:

$$\tilde{D} = \mathrm{sign}(\widehat{D})\ \max\big(0,\ |\widehat{D}| - \nu^2 \mathrm{SE}^2/|\widehat{D}|\big), \tag{7}$$

with dead zone $\{|r| \le \nu\}$; $\nu_{\mathrm{dev}} = 1.28$, $\nu_{\mathrm{LF}} = 2.33$. With $\nu = 1$, (7) is exactly the empirical-Bayes posterior mean under a $\mathcal{N}(0, \tau^2)$ prior with plug-in $\hat{\tau}^2 = \max(0, \widehat{D}^2 - \mathrm{SE}^2)$ [20]. The map is odd, continuous, never exceeds $|\widehat{D}|$, and is monotone in $\widehat{D}$ and anti-monotone in SE.

Link. Model $\varphi \,|\, y \sim \mathcal{N}(\mu_y, s^2)$ within item with the mixture standardized to unit variance, which is what (2) enforces, so $s^2 = 1/(1 + \bar{p}(1-\bar{p})u^2)$ where $u = \sqrt{2}\,\Phi^{-1}\big(\frac{1+\tilde{D}}{2}\big)$ and $\bar{p}$ is the base rate of correct traces. The Bayes-optimal per-trace log-weight coefficient is then

$$\gamma^* = \frac{u}{s} = u\sqrt{1 + \bar{p}(1-\bar{p})\, u^2}, \tag{8}$$

capped at $\gamma_{\max}$ (4 dev, 2 label-free). The uncorrected link $\gamma = u$ under-weights strong channels by up to ~50% at $D = 0.9$. The link assumes φ is within-item normal, which (2) supplies only asymptotically: at m_q=4 the score takes four values before standardization. The scale consequence is handled by using the realized σ_q, but the distributional one is not, and it bites hardest in the small-budget setting this paper advertises (m=40 estimates transferring to m=8). Where $\widehat{D}$ saturates the cap binds and the link's shape is irrelevant; small m is where it has to hold and where it is least justified.

D. tact in one expression

Two simplifications collapse the pipeline. Factoring $\widehat{D}$ out of (7) makes the shrinkage a multiplicative gain in the pooled z-statistic $\zeta = \widehat{D}/\mathrm{SE}$ alone, and substituting $u = \sqrt{2}\,z$ with $z = \Phi^{-1}\big(\frac{1+\tilde{D}}{2}\big)$ into (8) removes the nested radical. tact is then

$$\boxed{\begin{aligned} \hat{a}_q &= \arg\max_A \sum_{i:\, a_{q,i}=A} \exp\big(\gamma\, \varphi_{q,i}\big), \\ \gamma &= \Big[z\sqrt{2 + 4\bar{p}(1-\bar{p})z^2}\Big]_{-\gamma_{\max}}^{\gamma_{\max}}, \end{aligned}} \tag{9}$$

$$z = \Phi^{-1}\Big(\tfrac{1}{2}\big[1 + \widehat{D}\,(1 - \nu^2/\zeta^2)_+\big]\Big), \qquad \zeta = \tfrac{\widehat{D}}{\mathrm{SE}}, \tag{10}$$

with $\widehat{D}$ from (5) and φ from (2). At the default $\bar{p} = \frac{1}{2}$ the exponent is exactly

$$\gamma = z\sqrt{2 + z^2}, \qquad z = \Phi^{-1}\big(\tfrac{1+\tilde{D}}{2}\big), \tag{11}$$

Algorithm 1 tact: derive the exponent, then vote

Require: labeled dev pools $\mathcal{D}$, test pools $\mathcal{T}$, budget K, floor ν, cap $\gamma_{\max}$
Ensure: one scalar γ; an answer $\hat{a}_q$ for each $q \in \mathcal{T}$

$\mathcal{S} \leftarrow \emptyset,\;\; H \leftarrow \emptyset$
for $q \in \mathcal{D}$ **do**
  $y_i \leftarrow \mathbf{1}[a_{q,i} = a^*_q],\;\; i \le K; \quad H \leftarrow H \cup \{y\}$
  if $0 < \sum_i y_i < K$ **then** ▷ informative items only
    $R \leftarrow$ within-item midranks of $c_{q,1:K}$
    $D_q \leftarrow 2U_q/(n^1_q n^0_q) - 1, \quad U_q$ from R
    $\mathcal{S} \leftarrow \mathcal{S} \cup \{(D_q,\, N_q,\, \mathrm{Var}_0(D_q))\}$
$\widehat{D} \leftarrow \sum_q N_q D_q \,/\, \sum_q N_q$ ▷ van Elteren
$\mathrm{SE} \leftarrow \max\{\mathrm{SE}_0,\;\mathrm{SE}_J,\;1/(2\sqrt{N})\}$
$\bar{p} \leftarrow \mathrm{clip}(\mathrm{mean}(H),\, 0.05,\, 0.95)$
$\zeta \leftarrow \widehat{D}/\mathrm{SE}$
if $|\zeta| \le \nu$ **then** ▷ dead zone
  $\gamma \leftarrow 0$
else
  $\tilde{D} \leftarrow \widehat{D}\,(1 - \nu^2/\zeta^2)$ ▷ positive-part JS
  $z \leftarrow \Phi^{-1}\big((1 + \tilde{D})/2\big)$
  $\gamma \leftarrow \mathrm{clip}\big(z\sqrt{2 + 4\bar{p}(1-\bar{p})z^2},\, \pm\gamma_{\max}\big)$
for $q \in \mathcal{T}$ **do**
  if $\gamma = 0$ **then**
    $\hat{a}_q \leftarrow \mathrm{SC}(a_{q,1:K})$ ▷ same routine, bit-identical
  else
    $\varphi \leftarrow$ standardized van der Waerden scores of $c_{q,1:K}$
    $\hat{a}_q \leftarrow \arg\max_A \sum_{i: a_{q,i} = A} e^{\gamma\varphi_i}$
return γ, $\{\hat{a}_q\}$

one probit and one square root, where $\tilde{D}$ is the shrunk pooled statistic of (7) and not the per-item AUC_q of (5). Nothing in it is fitted to outcomes: ν is a significance level and $\gamma_{\max}$ a clip, both fixed before any data is seen. The clip is not cosmetic, though. Where $\widehat{D}$ saturates it binds, and the vote then sees $\gamma_{\max}$ rather than the derived magnitude (Section VIII). The dead zone is now visible as a single condition, $|\zeta| \le \nu$, on which γ is identically zero and (9) is bitwise sc by Proposition 1. Equations (9)–(11) are verified equivalent to the shipped implementation over randomised inputs including every boundary (tests/test_formula.py).

Algorithm 1 states the labeled path end to end. Both loops cost $O(K \log K)$ per item, dominated by the within-item ranking, and the estimate is a single scalar: nothing item-specific crosses from dev to test, which is what makes the dead zone a global abstention rather than a per-item one.

E. Anchor properties

Proposition 1 (Exact sc reduction). At $\gamma = 0$, (3) equals plain sc as a function on every trace pool, including tie-breaks. Under $D = 0$, $P(\gamma = 0) \to 2\Phi(\nu) - 1$ (80% dev, 98% label-free), and γ is continuous through the dead-zone boundary, so a false positive applies an infinitesimal exponent.

Algorithm 2 tact-LF: recovering the sign without labels

Require: pools $\mathcal{P}$, budget K, dedup threshold θ=0.95, margin quantile β=0.40, floor ν_{LF}, cap $\gamma_{\max}$, splits J=20, attenuation floor 0.20, minGated
Ensure: γ, equal to 0 whenever any alarm fires

1: **for** $q \in \mathcal{P}$ **do**
2:   $w_i \leftarrow 1/|\mathrm{group}(i)|$ from single linkage at dup $\ge \theta$
3:   $M_q \leftarrow \arg\max_A \sum_{i: a_{q,i} = A} w_i$ ▷ dedup-weighted plurality
4:   $\mathrm{mgn}_q \leftarrow$ top-two dedup-weighted share gap
5: $G \leftarrow \{q : \mathrm{mgn}_q \ge Q_\beta(\mathrm{mgn}),\; \ge 2$ distinct answers$\}$
6: $E_1 \leftarrow [\ \mathrm{median}_q(\text{Kish ratio}) < 0.5\ ]$ ▷ duplicate collapse
7: $E_4 \leftarrow [\ |G| < \text{minGated}\ ]$
8: $g_{q,i} \leftarrow \mathbf{1}[a_{q,i} = M_q]$ for $q \in G$
9: $(\widehat{D}_g, \mathrm{SE}_g, z_g) \leftarrow$ pooled statistic over G using g
10: $s \leftarrow \mathrm{sign}(\widehat{D}_g)$ ▷ estimated trust direction
11: $E_2 \leftarrow [\ \psi(s) > 0.05\ ]$ ▷ sign-aware margin decoupling
12: $\alpha \leftarrow$ mean two-half plurality agreement over J splits
13: $k \leftarrow$ inverse-Simpson size of the non-plurality mass
14: $p \leftarrow \big[1 + \sqrt{1 - (k{+}1)(1 - k\alpha)}\,\big]/(k{+}1)$
15: $E_3 \leftarrow [\ \text{discriminant} < 0.02\ ]$ ▷ root ambiguity
16: $\hat{\rho} \leftarrow \mathrm{clip}\big(\mathrm{UCB}_{95}(2p - 1),\, 0.20,\, 1\big)$
17: **if** $E_1 \vee E_2 \vee E_3 \vee E_4$ or $|z_g| \le \nu_{\mathrm{LF}}$ **then**
18:   **return** 0 ▷ refuse; the vote stays sc
19: **return** $\mathrm{Temper}\big(\widehat{D}_g/\hat{\rho},\; \mathrm{SE}_g/\hat{\rho}\big)$ with $\bar{p}$ unset

Proposition 2 (Exact CISC reduction). With the feature map $\varphi^{\log}_{q,i} = \log c_{q,i} - \overline{\log c_q}$, the weights equal $\lambda_q\, c^{\gamma}_{q,i}$ with a per-item constant $\lambda_q > 0$ (distinct from the coupling κ of (1)); hence the argmax, the ties, and the normalized vote shares coincide with CISC-power(γ) on every pool.

Proposition 3 (Regularity). The composite $g(\widehat{D}, \mathrm{SE})$ is continuous, odd, nondecreasing in $\widehat{D}$, nonincreasing in SE in magnitude, with $g(D, 0^+) = \gamma^*(D)$.

Proofs are elementary and pinned by unit tests in the released code (102 tests; the permutation-verified null variance, the EB identity in (7), and the link derivation (8) are each tested numerically).

V. Label-Free Estimation

A. Pipeline

(i) Dedup: single-linkage duplicate groups on the lexical-similarity channel at 0.95; each trace gets weight 1/|group| for plurality determination and pair weighting. (ii) Pseudo-label: $g_{q,i} = \mathbf{1}[a_{q,i} = M_q]$ with M_q the dedup-weighted plurality. (iii) Margin gate: keep the top 60% of items by dedup-weighted margin. (iv) Compute (5) with lab $= g$, giving $(\widehat{D}_g, \mathrm{SE}_g, r_g)$.

Two orderings in Algorithm 2 are load-bearing. The significance gate on line 17 tests the raw z_g, whose sign is unbiased by Proposition 4, while the tempering on line 19 uses the de-attenuated pair; testing the inflated statistic instead would let the de-attenuation manufacture significance. And $\bar{p}$ is left unset on the label-free path, so the mixture correction of (8) is not applied there: the base rate is exactly what no label-free estimator knows.

B. Sign consistency and its boundary

Proposition 4 (Attenuation identity). Let $\bar{\rho}$ be the pair-weighted probability that an item's plurality is wrong. If the plurality-error event is independent of φ given y (class-conditional noise), then $\mathbb{E}[\widehat{D}_g] = (1-2\bar{\rho})\,D$. In particular $\operatorname{sign}\mathbb{E}[\widehat{D}_g] = \operatorname{sign} D$ whenever $\bar{\rho} < 1/2$: the label-free estimate can only under-trust, never mis-sign.

The identity fails when the flip is caused by confidence, that is, under a confident echo. There the observable law under {majority right, $D < 0$} and {majority wrong via confident echo, $D > 0$} is identical (the two-root ambiguity of [10] restated for a single channel), so any label-free guarantee is necessarily conditional; it is stated as such rather than papered over.

C. De-attenuation and alarms

Split-half agreement over R=20 random half-splits estimates $\alpha = p^2 + (1-p)^2/k$ under a one-coin model with k effective wrong alternatives (inverse-Simpson), inverted as $p = [1 + \sqrt{1-(k+1)(1-k\alpha)}]/(k+1)$; $\widehat{D}_g$ is divided by the upper 95% bootstrap bound of $2p-1$ (floored at 0.2), which can only under-inflate. Four alarms force $\gamma = 0$: duplicate collapse (median Kish ratio < 0.5), sign-aware margin-decoupling, root ambiguity in the split-half quadratic, and insufficient gated items. The margin-decoupling alarm must condition on the estimated trust direction: a sign-naive version ("plurality has the highest mean φ") false-alarms on every benign anti-correlated channel—a defect encountered, diagnosed, and fixed during development, and pinned by the released tests. Finally the significance gate acts on the raw z (unbiased sign under Proposition 4) and temper on the de-attenuated value. A semi-label-free mode takes only the sign from ∼50 dev labels, routing it into the pipeline and disabling only the proxy-sign alarm; this purchases immunity to the ambiguity above at negligible labeling cost.

VI. Heterogeneity: Impossibility and Escape

A. Per-item adaptation is closed under i.i.d. coupling

Suppose $\kappa_q \overset{\text{iid}}{\sim} \mathcal{N}(0, 0.6^2)$ with no observable covariate.

Proposition 5 (Self-reinforcement). Any per-item rule $\gamma_q = h(\widehat{D}_q^g)$ with h monotone increasing and odd reinforces the plurality on both branches: $\widehat{D}_q^g > 0$ up-weights confident traces, which agree with the plurality; $\widehat{D}_q^g < 0$ up-weights unconfident traces, which are again the plurality side.

Remark 1. Measured in this harness, such a rule agrees with sc on 97.5% of items and its residual flips are net-harmful (1 right vs. 9 wrong per 400 items).

Remark 2 (Winner's curse). This is a measurement in the present harness rather than a theorem: on plurality-wrong items with $|D_q| > 0.3$, the items where a flip could win, the agreement statistic's sign matches the true sign only 4% of the time.

Proposition 6 (Two-world unidentifiability). Let $w_1 = \{\kappa > 0,\ \text{minority correct}\}$ and $w_2 = \{\kappa < 0,\ \text{plurality correct}\}$. Computed against either truth, $D^{w_1} = -D^{w_2}$, so the statistic tact uses cannot order the two worlds. When the item has exactly two answer clusters the laws of (a, c) coincide outright and no label-free method can separate them. With three or more populated clusters they do not coincide: under w_1 only the correct minority carries elevated confidence, whereas under w_2 every non-plurality cluster does, so the conditional law of c on a third cluster separates them (verified numerically; the two laws agree on the top two clusters and differ with KS $p < 10^{-15}$ on the third). The impossibility is therefore conditional on the pool being effectively binary, which is the regime the confident-echo cell occupies: 88% of its items have no third cluster at K=15. tact does not exploit the residual signal, and no published method does either; doing so is left open.

Consequently the per-item oracle (0.973 in this harness) is unreachable, and the honest behaviour is to fall back to the global estimate, which tact's dead zone does: in the i.i.d. cell every variant returns bitwise sc (zero discordant pairs).

B. TACT-group

Real heterogeneity is typically indexed by an observable covariate (domain, question type). With κ indexed by a group label, running the estimator per group keeps every group inside the operating regime of Sections IV–V; groups with fewer than 30 dev (or 60 unlabeled) items fall back to the global estimate, which Propositions 5–6 show is the only defensible default.

VII. Experimental Setup

Harness. A cluster-mixture oracle generates, per item, up to $K_{\max}$=20 cached traces with answers, confidences (1), and two similarity channels; all methods replay identical pools (paired comparisons, exact McNemar tests). Voting budget K=15; 400 items per cell on the sweep, 600 for the group study; dev splits of 200 (primary) and 50 (small-dev).

Regimes. The κ sweep $\{-0.6, \ldots, +0.6\}$; three strictly monotone confidence distortions (compression toward 0.5, over-confident sigmoid, fourth power), rank-preserving by construction, so discrimination is intact while calibration is destroyed; i.i.d. heterogeneity ($\kappa_q \sim \mathcal{N}(0, 0.6^2)$); covariate-structured heterogeneity (three groups at

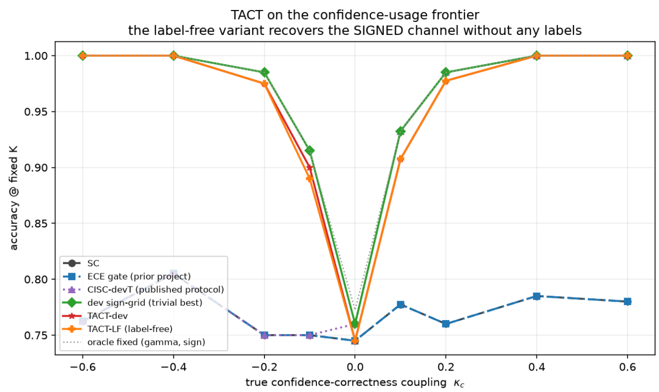

Fig. 2. Main result on the confidence-usage frontier. tact-dev and the fully label-free tact-LF track the signed oracle across the sweep; CISC-devT and the ECE gate sit at the sc floor for all $\kappa < 0$.

$+0.6/0/-0.6$); and a confident-echo poison (a wrong cluster echoes verbatim with confidence 0.95).

Baselines. sc; CISC-power with $\gamma \in \{0.25, \ldots, 4\}$; CISC-devT, the published dev-calibrated protocol (positive grid picked on dev); a binary ECE gate; SignGrid-dev, the strongest trivial baseline (signed exponent grid picked on dev); and the test-set oracle over signed fixed exponents as the upper envelope. The group study adds the naive self-referential per-item method as a negative control and the per-item link oracle as the ceiling.

Pre-registered falsifiers. The decision rule is stated here because the protocol is offered as a contribution. Each falsifier is an exact paired McNemar test on the 400 items of a cell, at $\alpha = 0.05$ one-sided, with the seed-level bootstrap of Section VIII-F as the second gate: a falsifier fires when the single-cell test is significant and the across-seed interval excludes zero. Single-cell tests at this size are underpowered against a strong baseline, which is the reason for the second gate; the earlier fixed $\tau = 0.02$ tolerance is retained only as a reporting convenience in the tables. F1: tact-dev below the best fixed-γ CISC at κ=+0.6 by more than τ. F2: either variant below sc by more than τ anywhere on the sweep. F3: the label-free variant fails to beat the ECE gate on sweep average. F4: CISC-devT or SignGrid-dev within τ of tact-dev everywhere, including the distortion, heterogeneity, and small-dev cells. F4 is the falsifier this matters most for: on a single cell the tact-minus-SignGrid comparison never reaches significance anywhere on the sweep (smallest $p = 0.08$ at $\kappa = +0.1$), and it is the ten-seed bootstrap that resolves the mid-range gap as systematic. A protocol that had reported only the single-seed tests would have called the mid-range a tie.

VIII. Results

A. Signed recovery, with and without labels

Table I and Fig. 2 give the sweep. Three observations. First, the published protocols never leave the floor on

TABLE I
Coupling sweep (accuracy at K=15; 400 paired items per cell; dev n=200). Published protocols sit at the sc floor on the entire negative half-axis.

κ	sc	ECE	devT	SignGrid	tact-dev	tact-LF	oracle
−0.6	.762	.762	.762	1.000	1.000	1.000	1.000
−0.4	.805	.805	.805	1.000	1.000	1.000	1.000
−0.2	.750	.750	.750	.985	.975	.975	.985
−0.1	.750	.750	.750	.915	.900	.890	.915
0.0	.745	.745	.760	.760	.745	.745	.772
+0.1	.777	.777	.932	.932	.907	.907	.932
+0.2	.760	.760	.985	.985	.978	.978	.985
+0.4	.785	.785	1.000	1.000	1.000	1.000	1.000
+0.6	.780	.780	1.000	1.000	1.000	1.000	1.000

TABLE II
Adversarial regimes (accuracy at K=15). "Oracle" is the test-set best over raw-value weight policies; rank invariance beats that entire family under compression.

Regime	sc	devT	SignGrid	tact-dev	tact-LF
Monotone compress	.775	.963	.963	1.000	1.000
Monotone overconf	.775	1.000	1.000	1.000	1.000
Monotone power	.775	1.000	1.000	1.000	1.000
Hetero (i.i.d.)	.765	.765	.765	.765	.765
Confident echo	.190	.190	.568	.615	.190[†]

[†]alarm fires and the method refuses to leave sc: the conditional guarantee of Prop. 4 working as stated.

$\kappa < 0$: CISC-devT's grid is positive-only and the ECE gate never opens (dev ECE ranges 0.10–0.80 across the sweep while the signal's discrimination is perfect at the extremes). Second, the label-free variant matches the 200-label variant nearly point-for-point—at κ=−0.6 the raw agreement statistic is $\widehat{D}_g = -0.81$ with $z = -17.6$, and the CCN identity's sign guarantee holds as predicted, yielding 1.000 with zero labels. Third, at $\kappa = 0$ the dead zone returns $\gamma = 0$ exactly, so the paired accuracy difference to sc is identically zero—"non-inferior" is replaced by "identical."

Where the derived exponent actually operates. The four cells that carry the headline 1.000 ($\kappa = \pm 0.4, \pm 0.6$) are cells where $\widehat{D}$ saturates, so the link returns an untempered γ^* between −8.4 and +12.1 across the seven saturated cells and the cap $\gamma_{\max} = 4$ is what the vote actually sees; the derived magnitude is not doing the work there, the sign is. Conversely, at $\kappa = \pm 0.1, \pm 0.2$, where the derived value lands strictly inside the cap ($|\gamma|$ from 1.10 to 2.91), tact-dev trails the dev-picked signed grid on all four cells (0.900 vs. 0.915; 0.907 vs. 0.932; 0.975 vs. 0.985; 0.978 vs. 0.985). Read together: against the published protocols the advantage is large and comes from representing the sign at all, whereas against a signed grid the analytic map is not better at choosing a magnitude on these cells. Its advantage over the grid is elsewhere, in the three cells named in C4, and the one place the interpolation itself pays is confident echo, where $\gamma = -1.198$ falls between grid points and beats the grid optimum $\gamma = -1$ (0.615 vs.

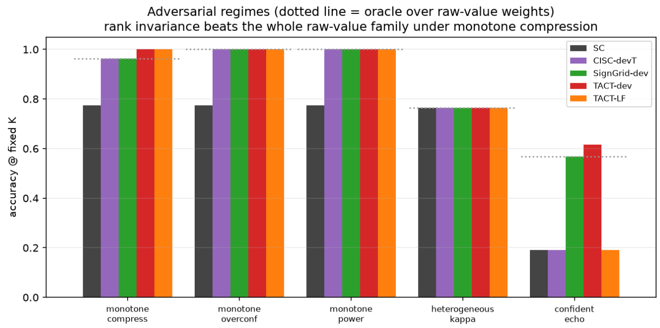

Fig. 3. Adversarial regimes. Dotted line: the oracle over raw-value weights. Left group of bars: rank invariance beats that family under compression; right: the labeled variant counters the confident echo while the label-free variant alarms and refuses.

0.568).

B. Rank invariance where raw values fail

Under monotone compression (Table II, Fig. 3) all confidences huddle near 0.5, so every c^{γ}-family weight is nearly uniform: even the oracle over raw-value policies reaches only 0.963. tact's rank scores are untouched by the distortion and both variants reach 1.000. Under the confident echo, dev labels reveal the inversion (high confidence $\Rightarrow$ wrong) and tact-dev counters with $\gamma = -1.20$, the best result in the field (0.615; 3.2$\times$ the sc floor); label-free, the duplicate-collapse alarm fires and the method correctly refuses—by Proposition 6 no label-free method could do better than a coin flip on the sign here, since 88% of the cell's items are effectively binary and the escape the proposition identifies is unavailable on them; pretending otherwise would be the real failure.

C. Heterogeneity

Table III and Fig. 4 give the group study. In the covariate-structured cell, per-group tact recovers each group's signed coupling (dev $\{+4.0, 0.0, -4.0\}$, label-free $\{+2.0, 0.0, -2.0\}$, the κ=0 group correctly dead-zoned—and cracks the floor that provably binds every global policy: the label-free variant reaches 0.923, within 0.023 of the per-item link oracle, with zero paired losses to sc over 600 items ($+83/-0$, $p = 2.1\times10^{-25}$). In the i.i.d. cell every method sits at the floor with zero discordant pairs, the naive self-referential control included: it cannot beat the plurality it is derived from. That control is the empirical face of Propositions 5–6, and the grouped cell is where it shows: given the same covariate the per-group estimator exploits, it reaches 0.787 against the 0.785 floor, two items in 600. The two arms land within seed noise of each other here (0.923 and 0.927 on one seed; 0.929$\pm$0.015 each over five). A natural explanation would be the arms' different exponent caps; the ablation rules it out. Sweeping the cap over $\{1, 2, 3, 4, 6, 8\}$ moves neither arm at all (spread 0.0000 for both), so the cap is not load-bearing in this cell and the difference is sampling variation.

TABLE III
Heterogeneity study (600 paired items; K=15).

Method	Grouped	i.i.d.
sc (floor)	.785	.752
tact global (dev)	.785	.752
tact-group (dev)	.927	.752
tact-group (label-free)	.923	.752
Naive per-item (neg. control)	.787	.752
Per-item link oracle (ceiling)	.947	.973

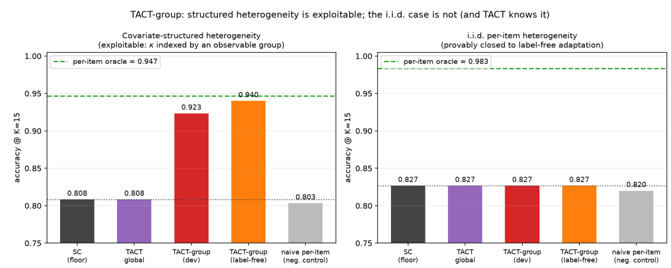

Fig. 4. Structured vs. i.i.d. heterogeneity. Left: with an observable covariate, per-group tact (label-free) approaches the per-item oracle from the 0.785 floor with zero losses to sc. Right: the provably closed i.i.d. cell—every method at the floor, the negative control included.

D. Small dev sets and falsifiers

With dev n=50 the conclusions are unchanged (1.000 at $|\kappa|$=0.6; 0.978 at -0.2): the SE-aware shrinkage degrades smoothly rather than catastrophically. All four falsifiers survived: F1 (1.000 vs. 1.000), F2 (bit-identical at κ=0; nowhere more than the pre-registered 0.02 accuracy tolerance below sc), F3 (sweep means 0.944 vs. 0.768), and F4 (both grid baselines trail by 0.037 and 0.047 on the distortion and echo cells, and neither can operate without labels). On the single seed the paired tests against SignGrid-dev are not significant anywhere ($\kappa{=}-0.2$: 3/7 discordant, exact $p = 0.34$; $\kappa{=}+0.1$: 8/18, $p = 0.08$), and the ten-seed intervals of Section VIII-F are what establish the mid-range deficit. Against SignGrid-dev the honest margin is narrow on the homogeneous sweep—tact even trails by 0.007–0.025 in the mid-range, the deliberate cost of shrinkage—and the net advantage concentrates exactly where pre-registered: distortion (+0.037), echo (+0.047), and label-free operation, which no grid can perform.

E. Verification of the implementation

Because every claim in Sections IV–VI is a mathematical property rather than an empirical trend, the released code pins each one with an executable test; the suite is 102 tests covering tact and the follow-on work. Table IV maps propositions to the tests that would fail if they stopped holding.

Two entries deserve comment. The permutation-invariance test was added after a defect in which the memoisation key made the test pass while the estimator itself was order-dependent by up to 0.10; it now calls the internal routine directly. And the last two rows are

TABLE IV

What the test suite verifies. Every proposition in the paper has an executable counterpart; the counter-tests fail deliberately on rejected alternatives so a regression cannot silently reinstate them.

Claim	Evidence
Prop. 1 (exact sc)	identical incl. ties, 200 pools; dead-zone rate >70% under D=0
Prop. 2 (exact CISC)	identical vote shares, 100 pools
Prop. 4 (attenuation)	$\rho \in \{.1, .25, .4\}$, abs. .06
Props. 5–6	97.5% sc agreement; 4% sign match; the two-world boundary is pinned both ways (binary pools indistinguishable, a populated third cluster separates them at KS $p < 10^{-10}$)
Rank invariance	3 distortions × 100 pools
Estimator internals	permutation null variance (3,000 draws, 10% tol.), JS–EB identity to 10^{-12}, link (8) to rel. 10^{-9}, permutation-invariance regression
Rejected alternatives	Kish ESS and the SAFE-under-VoI guarantee each have a test asserting their failure

counter-tests that assert failure of rejected alternatives — the Kish effective-sample-size formulation and the claim that the shipped default honours the SAFE stopping guarantee — so that neither can be silently reinstated by a later change.

F. A harness artifact, and dispersion across seeds

Two corrections to the synthetic results, both found by re-running what had been single-seed measurements.

Tie-breaking was rewarding sc for free. The generator assigned the correct answer the code 0 on every item, and argmax breaks ties toward the lowest index, so on any item whose vote was tied plain sc chose correctly by construction: 67.7% on near-tied items against 50.4% for a random tie-break, with 31.8% of items tied at K=15. Every method that perturbs the weights off integers forfeits that subsidy, so the artifact inflated the baseline and penalised the proposed method. It also produced a spurious falsifier: at $\kappa = 0$, tact-dev averaged 0.797 against sc at 0.818 across ten seeds, tripping F2, while the exponents responsible were $|\gamma| \leq 0.05$. The items were near-ties whose tie-break had moved, not items the exponent had reweighted. Answer codes are now permuted per item. With the artifact removed the two coincide exactly where the dead zone should hold: $\kappa = 0$ gives 0.745 for both methods and heterogeneous-κ gives 0.765 for both (ten-seed means 0.758 and 0.768, zero discordant pairs in every seed): the dead zone behaves as Proposition 1 states. All synthetic numbers in this paper are post-fix.

Dispersion. Ten seeds per cell, 400 paired items each, bootstrap over seeds. The extremes are stable to the third decimal (1.000 ± 0.001 at $|\kappa| \geq 0.4$). The mid-range deficit against SignGrid-dev is small but real rather than noise: -0.013 $[-0.017, -0.009]$ at $\kappa = -0.2$, -0.016 $[-0.024, -0.009]$ at -0.1, -0.012 $[-0.018, -0.005]$ at $+0.1$, -0.013 $[-0.019, -0.008]$ at $+0.2$, all $p < 0.001$. So is the advantage where the paper claims it: $+0.032$ $[+0.027, +0.037]$ on monotone compression and $+0.048$ $[+0.037, +0.059]$ on confident echo. The one cell that changes sign under the fix is heterogeneous-κ, now $+0.011$ $[0.000, +0.031]$ rather than a loss. Reporting a single seed would have hidden both the artifact and the fact that the mid-range gap is systematic.

G. Real-trace validation

Validation on real traces used Claude Haiku 4.5 as the frozen model: 100 items, 50 from GSM8K [23] and 50 from CommonsenseQA, with 12 independent chain-of-thought traces and a verbalized confidence per item (1,200 traces total), evaluated at K=12 with a 40/60 dev/test split. Four findings.

(a) The calibration–discrimination distinction reverses on real data, and tact reads it correctly. The channel is extremely well calibrated in the usual sense: ECE = 0.016, far inside the 0.10 gate, so a binary ECE gate opens and hands the channel to CISC. Yet the measured within-item discrimination is $\widehat{D} = -0.219$ with SE = 0.176 ($z = -1.24$): no usable signal, and what little there is points the wrong way (math -0.515, commonsense -0.173; both groups negative). This is the exact mirror image of the synthetic case in which ECE wrongly closed the gate on a discriminative channel (Section III): on real traces ECE wrongly opens it on a non-discriminative one. Calibration is uninformative about voting utility in both directions, and a signed discrimination statistic is what distinguishes them.

(b) The dead zone fires, and costs exactly nothing. With $|z| < \nu$, tact-dev, tact-LF and tact-group all return $\gamma = 0$ and are bit-identical to sc on every test item ($+0/-0$ discordant pairs, $p = 1$). All methods score 0.917. This is the pre-registered null-direction prediction of Section IX confirmed on real data: where the channel carries no signal, the method is free.

(c) Saturation is the binding constraint, not the estimator. Trace-level accuracy is 0.958 on GSM8K and 0.847 on CommonsenseQA, so only 12 of 100 items contain both a correct and an incorrect trace, the only items a within-item rank statistic can use. The estimator is not underpowered by design; the benchmark simply does not present the model with enough genuine uncertainty. Exposing non-null coupling on a strong model requires harder item pools, not more traces per item.

(d) Verbalized confidence is tie-heavy. Two values (0.99, 0.95) account for 49% of all reports, activating the tie-safe degeneration path of (2) on many items.

Scope of this first campaign: one model, two benchmarks, K=12. It confirms the null-direction prediction and the calibration–discrimination argument, and it is not evidence that tact improves accuracy, since the channel carried no signal to exploit. Finding (c) predicts what to do about that, and Section VIII-H does it.

TABLE V
Confirmatory campaign, MATH level-5 evaluation set (89 items, K=16). Every method replays the same cached pools. The duplication channel is inert because no reasoning text was collected, so dedup-sc coincides with sc by construction.

Method	Accuracy	net vs. sc
sc / dedup-sc / CISC-linear	0.888	—
tact-LF	0.888	$+0/-0$
tact-semi-LF	0.888	$+0/-0$
best-single-confidence	0.843	$+1/-5$
in-pool oracle (ceiling)	0.933	$+4/-0$

H. Confirmatory campaign on harder items

Finding (c) predicts that a channel measured as null on saturated benchmarks should become measurable on items the model finds genuinely uncertain. A pre-registered follow-up tests that prediction: 119 MATH level-5 problems [24], [25], 16 traces each from the same frozen model, a 30-item sign set and an 89-item evaluation set drawn from the registered list before any trace was collected, and five hypotheses (H1–H5) fixed in advance.

The channel is real. On the evaluation set the pooled statistic is $\widehat{D} = +0.250$ with SE $= 0.098$, so $z = +2.54$ and H1 passes. This is the first real-trace evidence that verbalized confidence carries positive within-item discrimination; the same measurement on GSM8K/CommonsenseQA gave -0.219 ($z = -1.24$).

The endpoint was unpassable for any method. The realized substrate saturated again: per-trace accuracy 0.819, sc 0.888, a decisive stratum of 10 of 89 items, and the correct answer present in the pool on only 4 of those. The in-pool oracle therefore tops out at $+4/-0$, exact one-sided $p = 0.0625$. H2 fails, but it fails for every conceivable aggregation method including a perfect one, so the failure is a property of the substrate rather than of the estimator.

Abstention behaved as designed. tact returned $\gamma = 0$, with alarms E4 and E2 firing on the label-free path and the sign set holding too few informative items to supply a semi-label-free sign. The vote is therefore bit-identical to sc at 0.888 (H3, H4 pass). The cost of acting anyway is visible in the same table: best-single-confidence, the trivial baseline that always trusts the channel, loses 4.5 points at 0.843.

One caveat from this campaign transfers beyond tact. Measured difficulty depended on the collection protocol: a 30-problem-per-call probe put level-5 plurality accuracy at 0.40, while the 15-problem-per-call confirmatory run yielded 0.888 on the same stratum. Batch size belongs in the experimental record whenever traces are collected in batches.

I. How wide is the addressable stratum?

Both campaigns failed their endpoint for the same reason, which suggests measuring that reason directly. Define the window as the fraction of items where the plurality is wrong and the correct answer is present in the pool: the ceiling for any label-free aggregation method, since nothing outside it can be changed.

TABLE VI
The addressable stratum across substrates, on one definition throughout: decisive is the fraction of items whose plurality is wrong, window (Win.) the fraction that are decisive and have the correct answer somewhere in the pool; both in per cent. The window is the ceiling for any label-free aggregation method. Rows marked † are measured here; HumanEval+/MBPP+ is recomputed from published oracle-minus-selector tables, which report the window only. As items harden the decisive fraction grows but the window does not: they pass from saturated to capability-limited. The MATH L5 row is over the full registered list of 119 items, the substrate being the object measured; Section VIII-H quotes the same quantities over the 89-item evaluation set that remains after the sign set is split off (10 and 4 items, so 11.2 and 4.5).

Domain	Substrate	Dec.	Win.
QA	GSM8K / CSQA†	9.0	4.0
QA	MATH L5†	7.6	2.5
QA	AIME / AMC†	23.3	3.3
Code	HumanEval+ / MBPP+	—	3.6
Code	LeetCode Med/Hard†	25.0	7.5
QA, budget-capped	MATH L5†	18.5	11.8

The window was measured on five substrates spanning two domains (Table VI). For code generation, where an executable test suite supplies per-sample ground truth and the window might reasonably be expected to widen, 40 LeetCode Medium/Hard problems [26] were solved 8 times each and graded against the benchmark's hidden suites, with the baseline taken as the largest behavioural cluster over probe inputs (never expected outputs). The window is $3/40 = 7.5\%$ (CI_{95} 2.6–19.9%): wider than label-free QA, but the same order, and the composition is the same shape at 30 saturated, 7 capability wall and 3 rescuable, i.e. 75/17.5/7.5%. Nor does budget open it. The seven capability-wall problems produced zero correct solutions in 224 further attempts (per-problem 95% upper bound on the pass rate 0.088), and extrapolating oracle@N shows the window saturating by N=32.

One precaution belongs with these numbers, because omitting it would have inverted them. The grading harness was validated against the benchmark's own reference solutions before any candidate was scored: 178 of 180 pass under the sandbox's resource limits. The check is not a formality: a sandbox whose resource limits the host rejects outright fails 100% of executions, and that condition presents as a candidate failure rather than as an error. Studies that grade by execution should report their reference-solution pass rate for the same reason a calibration curve is reported: without it, a broken harness and a capability wall look identical.

J. A third axis: reasoning budget

Difficulty is not the only way to lower per-sample accuracy. Constraining the reasoning budget lowers it while leaving the correct answer inside the model's competence, which is the combination the decisive stratum needs and that difficulty does not supply: as items harden they pass from saturated to capability-limited without pausing in between. A paired run tests this on the same 119 items, same model, same K, with the model instructed to answer without writing any working.

The manipulation was weaker than pre-registered on its stated check (per-sample accuracy $0.853 \to 0.781$, against a 0.10 drop required) and the correct answer did not leave the pool ($0.950 \to 0.933$), so what follows is exploratory. It is reported because the constraint did not lower accuracy uniformly, it moved items across the plurality boundary: sc fell $0.924 \to 0.815$, the window widened from 2.5% to 11.8% (paired exact McNemar on window membership, 13 in and 2 out, $p = 0.0037$), and the channel grew stronger, $\widehat{D}$ from $+0.229$ ($z = 2.69$) to $+0.398$ ($z = 6.76$). Forced to answer without working, the model's confidence tracks whether it happened to be right.

This is the widest window measured anywhere in this paper, and the first substrate on which the in-pool oracle clears the endpoint at all ($+9/-0$, $p = 0.002$ on a decisive stratum of 17). tact nonetheless returned $\gamma = 0$: only 12 items survived the margin gate against the threshold of 30, so E4 fired. That abstention is testable rather than a matter of taste, and it cost nothing. Estimating γ from gold labels, an upper bound no deployable method has access to, the best available is $+4/-1$ at $\gamma = 1$ ($p = 0.19$); the derived $\gamma = 0.670$ gives $+3/-1$. The gap that matters is therefore not the one the window closes. Even with the window at 11.8% and an oracle able to convert nine items, confidence weighting reaches fewer than half of them, and none of the reachable configurations is significant.

IX. Discussion and Limitations

What the evidence does and does not show. The accuracy claims are all on a synthetic oracle whose confidence model (1) is, at the homogeneous cells, the very coupling the estimator measures. Three design choices limit the circularity, and the first is weaker than it looks: heterogeneity and echo lie outside the estimator's working model, but the three distortion cells are rank-preserving by construction and therefore sit inside tact's own invariance group, so passing them tests that the implementation respects an invariance it was built to have rather than probing an untested regime; mechanism-recovery claims (does $\widehat{D}$ track κ?) are reported separately from accuracy claims; and the pre-measured baseline landscape (Fig. 1) fixed the winnable cells before the method existed. The real-trace campaigns of Sections VIII-G and VIII-H test the premise and the abstention behaviour, and both predictions held: the channel is null on saturated benchmarks and positive on competition mathematics, and the dead zone kept the vote bit-identical to sc in each case. They do not test the accuracy claim, because on neither substrate was the addressable stratum large enough for any method to demonstrate a gain (Section VIII-I).

Narrow margins where labels abound. When labels are plentiful and the confidence scale is trusted, a dev-picked signed grid captures most of the value; tact's case rests on the label-free setting, distorted scales, small dev sets, and the exactness of its anchors.

Conditional label-free guarantee, and what happens past the boundary. Proposition 4 requires $\bar{\rho} < 1/2$ after deduplication, and the confident-echo ambiguity is fundamental (Proposition 6). Follow-on work measured the consequence of crossing that boundary, and it is worse than under-trust. In a paraphrased wrong-majority cell (a dominant wrong cluster that is semantically tight but carries no verbatim signature, so deduplication has nothing to collapse) the plurality is wrong on most items, $\bar{\rho} > 1/2$, and tact-LF does not merely shrink toward sc: it mis-signs, saturates at $\gamma = -2.0$, and scores 0.000 against an sc floor of 0.340. None of the four alarms fires, because E1 keys on verbatim duplication which is absent by construction. This is the method's sharpest unguarded failure mode: the guarantee is conditional, the condition is not observable label-free, and the existing diagnostics do not detect its violation. Where a systematically wrong majority is plausible, the semi-label-free mode (sign from $\sim$50 labels) should be the default rather than an optional refinement. The other standard remedy is not tried here and should be: the failure is driven by a semantically tight wrong cluster that lexical deduplication cannot see, which is precisely what semantic-equivalence clustering [27] is built to collapse. Substituting a semantic pseudo-label for the lexical one is the obvious next guard, and this paper does not test it.

Global exponent per group. Within a group, tact ships one exponent; per-item variation inside a group is unexploitable by Propositions 5–6 unless further covariates exist.

The thin window. Section VIII-I measures the stratum this whole family of methods can act on at 2.5–7.5% of items on every substrate tried, in two domains, with no widening as items harden: they pass from saturated straight to capability-limited. Two consequences follow for the method proposed here. First, abstention is not a conservative compromise but the only correct default, and the measured cost of acting anyway was negative on both real substrates (best-single-confidence loses 4.5 points in Table V where the dead zone holds tact at the sc floor). Second, an aggregation gain of the size reported on the synthetic harness is not measurable on a benchmark of a few hundred items at these window widths, which is why the real-trace claim in this paper is confined to the premise (the channel exists and is signed) and to the abstention behaviour, and does not extend to accuracy. Demonstrating the gain needs a (model, benchmark) pair whose plurality is wrong on 30–60% of items with the correct answer still reachable, and no pair tried here satisfies both.

X. Conclusion

tact turns "how far should this model's confidence be trusted?" into a measured, signed, uncertainty-aware scalar, and recovers the sign without labels under conditions this paper states and tests. The measurement that frames it matters more than the estimator: across five substrates in two domains, the stratum on which any label-free aggregation method can act is 2.5–7.5% of items, and on both real substrates the in-pool oracle cannot clear a pre-registered endpoint. In that regime the useful property of an estimator is knowing when not to act, which the dead zone does exactly.

Code and Data Availability

All code, cached traces, and the JSON artifacts behind every table are at https://github.com/vito1317/adaptive-reasoning-consensus. Table IV is produced by pytest at commit 35ad160; the synthetic results by experiments/run_tact_eval.py, the real-trace campaigns by run_tact_hard_eval.py, and the window measurements by run_g1_window.py and run_g1_deepening.py. Each script writes the artifact its table cites.

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